MILITARY SPECIFICATION

PRESS BRAKES, POWER OPERATED, HYDRAULIC, SHEET AND PLATE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers direct acting and mechanically linked, hydraulically operated, sheet and plate press brakes of the gap or open throat design in a range from 20 to 2000 ton capacity with equipment and accessories specified herein.

1.2 Classification. Press brakes shall be of the sizes as listed in table I. The sizes to be furnished shall be as specified (see 6.2.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specification and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3441
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-23T04:34Z
Check the source to verify that this is the current version before use.

SPECIFICATION

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.

2.1.2 Other Government publications. The following other Government publication forms a part of this specification to the extent specified herein.

U. S. DEPARTMENT OF LABOR

29 CFR, Chapter XVII, Part 1910 - Occupational, Safety, and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, and other Government publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B5.1 - T-Slots, Their Bolts, Nuts and Tongues.

ANSI B11.3 - Safety Requirements for the Construction, Care, and Use of Power Press Brakes.

Source: https://assist.dla.mil -- Downloaded: 202610-23T04:34Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

JOINT INDUSTRIAL COUNCIL (JIC)

JIC H-1 - Hydraulic Standards for Industrial Equipment and General Purpose Machine Tools.

(Applications for copies should be addressed to the Joint Industrial Council, 7901 Westpark Drive, McLean, VA 22101.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Machine Tools.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The press brake shall be new and one of the manufacturer's current models capable of forming, bending, flanging, and punching in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Dimensional systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:34Z
Check the source to verify that this is the current version before use.

In this specification, all dimensions, measurements, sizes, and capacities are given in U.S. units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, depth stops, dial indicators, pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.2 Dual calibrations. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and the SI System of Measurements. When a dual US and SI system is furnished, both dials shall have independent zero adjustments and both dials shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the machine's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The press brake and all its applicable components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements contained in 29 CFR, Chapter XVII, Part 1910, and ANSI B11.3 that are applicable to the machine itself. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4).

3.2.5 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:34Z
Check the source to verify that this is the current version before use.

3.2.6 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.7 Environmental protection. The machine shall be so designed and constructed that, under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at point of installation (see 6.4).

3.2.8 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a cleanable or replaceable filter. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible. When specified (see 6.2.1), a centralized automatic lubrication system shall be provided.

3.2.9 Interchangeability. To provide for replacement of worn or damaged parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

3.3 Construction. The press brake shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine at its maximum bending capacity as listed in table I without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects such as scale and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:34Z
Check the source to verify that this is the current version before use.

TABLE I. Sizes and capacities.

Size	Tonnage	Length of die surface, feet	Bending capacity length x thickness	Distance between housings	Throat depth, inches	Stroke length, inches	Closed height, inches	Open height, inches	Bed top width, inches	Motor H.P.
20-4	20	4	4' x 14 ga	3'2"	6	2	6	8	2	3
35-4		4	4' x 12 ga	3'2"						
35-6	35	6	6' x 14 ga	5'2"	6	2	6	8	2	4
35-8		8	8' x 16 ga	5'2"						
35-10		10	10' x 18 ga	5'2"						
55-6		6	6' x 10 ga	4'2"						
55-8	55	8	8' x 12 ga	6'2"	6	2	6	8	2	6
55-10		10	10' x 14 ga	8'2"						
90-8		8	8' x 10 ga	6'2"						
90-10	90	10	10' x 10 ga	8'2"	6	4	6	10	3	7
90-12		12	12' x 12 ga	10'2"						
100-6		6	6' x 3/16"	4'2"						
100-8	100	8	8' x 3/16"	6'2"	6	5	5	10	3	7
100-10		10	10' x 10 ga	8'2"						
100-12		12	12' x 12 ga	10'2"						
135-8		8	8' x 1/4"	6'2"						
135-10	135	10	10' x 3/16"	8'2"	7	5	5	10	3	10
135-12		12	12' x 10 ga	10'2"						
135-14		14	14' x 10 ga	12'4"						
150-8		8	8' x 1/4"	6'2"						
150-10	150	10	10' x 3/16"	8'2"	7	7	5	12	4	10
150-12		12	12' x 3/16"	10'2"						
150-14		14	14' x 10 ga	12'3"						
150-16		16	16' x 10 ga	14'4"						

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:34Z
Check the source to verify that this is the current version before use.

TABLE I. Sizes and capacities - Continued.

Size	Tonnage	Length of die surface, feet	Bending capacity length x thickness	Distance between housings	Throat depth, inches	Stroke length, inches	Closed height, inches	Open height, inches	Bed top width, inches	Motor H.P.
175-8		8	8' x 5/16"	6'2"						
175-10		10	10' x 1/4"	8'2"						
175-12	175	12	12' x 3/16"	10'2"	7	7	5	12	5	12
175-14		14	14' x 10 ga	12'4"						
175-16		16	16' x 12 ga	14'4"						
200-8		8	8' x 5/16"	6'4"						
200-10		10	10' x 1/4"	8'4"						
200-12	200	12	12' x 1/4"	10'4"	8	7	5	12	5	12
200-14		14	14' x 3/16"	12'4"						
200-16		16	16' x 3/16"	14'4"						
225-8		8	8' x 3/8"	6'4"						
225-10		10	10' x 5/16"	8'4"						
225-12	225	12	12' x 1/4"	10'4"	8	7	5	12	5	20
225-14		14	14' x 3/16"	12'4"						
225-16		16	16' x 3/16"	14'4"						
250-10		10	10' x 3/8"	8'4"						
250-12	250	12	12' x 5/16"	10'4"	8	7	5	12	5	20
250-14		14	14' x 1/4"	12'4"						
250-16		16	16' x 3/16"	14'4"						
300-8		8	8' x 1/2"	6'4"						
300-10		10	10' x 3/8"	8'4"						
300-12	300	12	12' x 5/16"	10'4"	8	7	6	13	5	25
300-14		14	14' x 5/16"	12'4"						
300-16		16	16' x 1/4"	14'4"						
300-18		18	18' x 3/16"	16'4"						

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:34Z
Check the source to verify that this is the current version before use.

TABLE I. Sizes and capacities - Continued.

Size	Tonnage	Length of die surface, feet	Bending capacity length x thickness	Distance between housings	Throat depth, inches	Stroke length, inches	Closed height, inches	Open height, inches	Bed top width, inches	Motor H.P.
350-8		8	8' x 5/8"	6'4"						
350-10		10	10' x 1/2"	8'4"						
350-12	350	12	12' x 1/2"	10'4"	10	8	8	14	5	25
350-14		14	14' x 3/8"	12'4"						
350-16		16	16' x 5/16"	14'4"						
350-18		18	18' x 1/4"	16'4"						
400-8		8	8' x 1/2"	6'4"						
400-10		10	10' x 1/2"	6'4"						
400-12		12	12' x 7/16"	8'4"						
400-14	400	14	14' x 3/8"	10'4"	10	8	8	14	7	30
400-16		16	16' x 5/16"	12'4"						
400-18		18	18' x 5/16"	14'4"						
400-20		20	20' x 1/4"	16'4"						
500-10		10	10' x 3/4"	6'4"						
500-12		12	12' x 5/8"	8'4"						
500-14		14	14' x 5/8"	10'4"						
500-16	500	16	16' x 1/2"	12'4"	10	8	8	16	8	30
500-18		18	18' x 3/8"	14'4"						
500-20		20	20' x 3/8"	16'4"						
500-24		24	24' x 5/16"	20'4"						
600-10		10	10' x 7/8"	8'4"						
600-12		12	12' x 3/4"	8'4"						
600-14		14	14' x 3/4"	10'4"						
600-16	600	16	16' x 1/2"	12'4"	13	12	8	20	8	40
600-18		18	18' x 3/8"	14'4"						
600-20		20	20' x 5/16"	16'4"						
600-22		22	22' x 1/4"	18'4"						
600-24		24	24' x 3/16"	20'4"						

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:34Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-02-23T04:34Z
Check the source to verify that this is the current version before use.

TABLE I. Sizes and capacities - Continued.

Size	Tonnage	Length of die surface, feet	Bending capacity length x thickness	Distance between housings	Throat depth, inches	Stroke length, inches	Closed height, inches	Open height, inches	Bed top width, inches	Motor H.P.
750-12		12	12' x 1"	8'4"						
750-14		14	14' x 1"	10'4"						
750-16		16	16' x 7/8"	12'4"						
750-18	750	18	18' x 5/8"	14'4"	12	12	10	22	10	40
750-20		20	20' x 1/2"	16'4"						
750-22		22	22' x 7/16"	18'4"						
750-24		24	24' x 3/8"	20'4"						
1000-12		12	12' x 1-1/4"	8'4"						
1000-14		14	14' x 1-1/4"	10'4"						
1000-16		16	16' x 1"	12'4"						
1000-18	1000	18	18' x 3/4"	14'4"	14	12	10	22	12	60
1000-20		20	20' x 3/4"	16'4"						
1000-22		22	22' x 5/8"	18'4"						
1000-24		24	24' x 1/2"	20'4"						
1250-12		12	12' x 1-3/8"	8'4"						
1250-14		14	14' x 1-1/4"	10'4"						
1250-16		16	16' x 1-1/8"	12'4"						
1250-18		18	18' x 1"	14'4"						
1250-20	1250	20	20' x 7/8"	16'4"	16	12	10	22	16	60
1250-22		22	22' x 3/4"	18'4"						
1250-24		24	24' x 5/8"	20'4"						
1250-26		26	26' x 1/2"	22'4"						
1250-28		28	28' x 3/8"	24'4"						

TABLE I. Sizes and capacities - Continued.

Size	Tonnage	Length of die surface, feet	Bending capacity length x thickness	Distance between housings	Throat depth, inches	Stroke length, inches	Closed height, inches	Open height, inches	Bed top width, inches	Motor H.P.
1500-12		12	12' x 1-7/8"	8'4"						
1500-14		14	14' x 1-3/4"	10'4"						
1500-16		16	16' x 1-1/2"	12'4"						
1500-18		18	18' x 1-1/4"	14'4"						
1500-20		20	20' x 1-1/4"	16'4"						
1500-22	1500	22	22' x 1-1/4"	18'4"	18	12	10	22	16	60
1500-24		24	24' x 1"	20'4"						
1500-26		26	26' x 7/8"	22'4"						
1500-28		28	28' x 7/8"	24'4"						
1500-30		30	30' x 3/4"	26'4"						
1500-32		32	32' x 3/4"	28'4"						
1500-34		34	34' x 5/8"	30'4"						
2000-12		12	12' x 2"	8'4"						
2000-14		14	14' x 1-7/8"	10'4"						
2000-16		16	16' x 1-3/4"	12'4"						
2000-18		18	18' x 1-1/2"	14'4"						
2000-20		20	20' x 1-1/2"	16'4"						
2000-22	2000	22	22' x 1-1/4"	18'4"	20	18	12	30	16	60
2000-24		24	24' x 1-1/4"	20'4"						
2000-26		26	26' x 1-1/4"	22'4"						
2000-28		28	28' x 1-1/4"	24'4"						
2000-30		30	30' x 1"	26'4"						
2000-32		32	32' x 7/8"	28'4"						
2000-34		34	34' x 3/4"	30'4"						

NOTE: Requirements are not less than those shown.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:34Z
Check the source to verify that this is the current version before use.

condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the press brake shall be painted in accordance with the manufacturer's commercial practice and color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28.

3.3.7 Gears. All gears shall be constructed from a material suitable for the intended purpose. All drive gears shall have the hardness, surface finish and toughness that will enable the gear train to transmit full rated torque of the drive motor without gear damage, failure, or premature deterioration and wear.

3.3.8 Electromagnetic interference control. When specified (see 6.2.1), equipment procured under this specification shall comply with the requirements of MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The press brake shall consist of not less than a frame, a bed plate, a ram, a hydraulic system, controls, backgauges, and an electrical system.

3.4.1 Frame. The frame shall be of the open throat design consisting of a bed plate and side housings constructed of welded steel plates. Contoured stress points shall be utilized to avoid concentration of stresses at sharp corners. The frame shall have adequate strength and rigidity to support all components and withstand all forces encountered when bending sheet and plate of full capacity length and thickness as specified in table I, applicable to the machine being procured. Mounting bolt holes shall be provided for fastening the machine to the floor.

3.4.1.1 Bed plate. The bed plate shall be a solid steel plate joined to the side frames by either the interlocked and bolted construction method or welded to the side frames and stress relieved. The top of the bed plate shall have a flat machined surface and shall be slotted the full length to accommodate the manufacturer's standard die holder, filler blocks, or die tongues. The bed top width shall be not less than the dimensions shown in table I. Means shall be provided for adjusting and securely locking dies or die holders to the bed. Press brakes of 200 ton capacity and larger shall have four or more T-slots conforming to ANSI 85.1, running at right angles to the tongue slot, for attaching short dies or fixtures. Means shall be provided for attaching gauges and fixtures to the bed. When specified (see 6.2.1), the front and rear face of the bed shall be machined and provided with a method of mounting wide angle support brackets or flanges. When specified (see 6.2.1), a die holder or filler block equal in length to the die surface specified in table I, as applicable to the machine being procured shall be furnished. The type of punch and die to be furnished shall be as specified and fully described (see 6.2.1).

Source: https://assist.dla.mil -- Downloaded: 202610-23T04:34Z
Check the source to verify that this is the current version before use.

3.4.2 Ram. The ram shall be of solid steel plate fitted with adjustable slides to work against steel or laminated self lubricating non-metallic ways or antifriction rollers running on hardend ways. The lower edge of the ram shall be slotted to receive the manufacturer's standard die tongues. A positive clamping device shall be incorporated to prevent die movement under load. When specified (see 6.2.1), the front and rear face of the ram shall be machined and provided with a method of mounting wide angle brackets or flanges.

3.4.2.1 Ram or bed tilt adjustment. Unless otherwise specified (see 6.2.1), the press brakes covered herein shall have means for tilting the ram or bed in either direction. The tilting device shall permit accurate adjustment of the desired angle between the ram and bed. An indicator to show the direction and amount of tilt in thousandths of an inch enabling the operator to return to any given setting shall be provided on all brakes. The tilt indicator shall be installed in a location readable from the operating position.

3.4.2.2 Ram leveling control. The ram leveling control shall be capable of maintaining ram level within plus or minus 0.002 inch while bending sheet or plate of the full length and thickness capacity as specified in table I for the applicable machine.

3.4.3 Hydraulic system. The hydraulic system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level. When specified (see 6.2.1), the hydraulic system shall conform to JIC H-1 Hydraulic Standards. The press brake shall be furnished with a three-speed system.

3.4.4. Ram hydraulic cylinder units. Unless otherwise specified, the press brake shall have either one or more hydraulic units with mechanical linkage for driving the ram, or two or more direct acting hydraulic cylinder units. When only one type unit is acceptable, the particular hydraulic unit required shall be as specified (see 6.2.1). The cylinder unit shall be solid one-piece, tie rod, or bolted construction. The hydraulic cylinder unit shall be made of iron or steel. Each ram hydraulic unit shall be the double action piston type and the piston rod of each cylinder shall be sealed against leakage. The ram end of the piston rod on the direct acting cylinder units shall be provided with ball sockets or aligning joints with an adjustment for length and wear. The hydraulic units with the mechanical linkage shall also have adjustment for wear.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:34Z
Check the source to verify that this is the current version before use.

3.4.5 Operating controls. The press brakes shall be furnished with the necessary controls for cycling or stopping the ram. When used in conjunction with the hydraulic system the controls shall provide at least a high speed advance, slow working stroke, and a high speed return of the ram. The controls shall be clearly marked and located conveniently to the operator at his normal work station.

3.4.5.1 Stroking control. A stroking control shall be furnished to select the different modes of operation. The operating modes shall include, at least the following:

a. Off. The press brake shall be inoperative.
b. Single stroke. The ram shall complete only one stroke regardless of how long the actuating switches are energized.
c. Inch. Allow jogging of the ram at any point within the stroke.

3.4.5.2 Actuator controls. Unless otherwise specified (see 6.2.1), the actuator controls shall consist of both a remote foot switch and dual palm buttons. Machines having both controls shall have a means of selecting hand or foot operation. The palm buttons shall be connected so that concurrent pressure from both hands is required to actuate the press brake. The palm buttons may be located on the bed, ram, or on a free standing pedestal and shall be protected against unintentional actuation.

3.4.5.3 Speed controls. When ram speeds other than those listed in 3.4.5 are desired the speed combinations required shall be as specified (see 6.2.1). The speed control furnished shall enable the operator to select the advance, press, and return speed combinations best suited for the job. Necessary cams and switches shall be provided to set the desired speed change points throughout the stroke range.

3.4.5.4 Stroke length controls. Controls shall be provided to adjust the stroke length. The top stop and bottom depth stop settings shall be adjustable. The depth stop shall be adjustable within plus or minus 0.001 inch.

3.4.5.5 Tonnage control. The press brake shall be equipped with a conveniently located adjustable tonnage control and tonnage indicator that shows the load at all times.

3.4.6 Back gauges. Unless otherwise specified, the press brake shall be equipped with a set of manually operated universal back gauges, of the manufacturer's standard design. When specified either a manual front operated back gauge or a powered front operated back gauge shall be furnished. The type back gauge required shall be as specified (see 6.2.1). The back gauge provided shall accommodate the thickest material for which the machine is designed and shall have a gauge position indicator graduated in not more than 0.010 inch increments. The indicator for the front operated gauges shall be readable from the operator's station.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:34Z
Check the source to verify that this is the current version before use.

3.4.7 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to NFPA 79. Each machine shall draw all of its electrical power from a single 230/460 volt, 3 phase, 60 Hz circuit. Each machine shall have a fused safety disconnect switch or circuit breaker. The machine shall be initially wired for operation on 460 volts. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in NFPA 79 for the disconnect fuse rating.

3.4.7.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2.1), each motor shall have a dripproof enclosure.

3.4.7.2 Solid state components. Solid state design shall be used throughout for electronic components.

3.4.7.3 Control circuit voltage. Auxiliary control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary voltage no greater than 120 volts.

3.5 Sizes and capacities. Unless otherwise specified (see 6.2.1), the size and capacity of the machine shall be not less than those specified in table I for the machine size specified. The bending capacities are based upon steel sheet and plate with a tensile strength of not less than 60,000 pounds per square inch (psi).

3.6 Performance. The press brake shall be capable of air bending the maximum length and thickness of steel sheet and plate into a 90 degree angle with one stroke of the ram over a v-die having an opening eight times the metal thickness for material not greater than 1/2 inch, and ten times metal thickness for material not less than 1/2 inch. The inside bend radius shall be 5/32 times the width of the v-die opening. The press brake shall be capable of forming angular and channel shapes, and multiple punching operations.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.7 Standard equipment. Unless otherwise specified, all equipment normally furnished by the manufacturer on his standard commercial machine shall be furnished including one charge of hydraulic fluid. When specified (see 6.2.1), required additional equipment shall be furnished and fully described.

Source: https://assist.dla.mil -- Downloaded: 202610-23T04:34Z
Check the source to verify that this is the current version before use.

3.8 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.9 Repair parts. Repair parts shall be furnished as specified (see 6.2.1).

3.10 Marking on plates and charts. All words on instructions, indicating plates, and nameplates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

3.10.1 Lubrication chart or plate. A lubrication chart or plate shall be permanently and securely attached to each machine. The chart shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.10.2 Nameplate. A nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11 Hour meter. Unless otherwise specified (see 6.2.1), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of one hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine. Upon reaching 99,999 hours, the meter readout shall automatically return to zero and continue to accumulate time.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:34Z
Check the source to verify that this is the current version before use.

3.12 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be written in the English language.

3.13 Workmanship. Workmanship of the press brake and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination and all tests shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 202610-23T04:34Z
Check the source to verify that this is the current version before use.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1) quality conformance inspection shall consist of at least the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, test and inspection shall be cause for rejection.

4.6 Examination. The machine shall be visually examined to determine compliance with the requirements of this specification.

4.7 Tests.

4.7.1 Operational test. The machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.2 Performance tests. The press brake shall be fitted with a punch and a v-die having an opening eight times metal thickness if the test piece is less than 1/2 inch thick; ten times metal thickness if the test piece is 1/2 inch or thicker. The inside bend radius shall be 5/32 times the width of the v-die opening. The punch and v-die shall be designed to compensate for material spring back. The test sheet or plate shall have a tensile strength of not less than 60,000 psi. The following tests shall be performed.

4.7.2.1 Full length bend test. The test piece shall be of the type and strength as stated in 4.7.2, having length and thickness equal to the maximum capacity of the machine as specified in table I. The test shall be accomplished by inserting the test piece between the dies and forming a 90 degree bend with one stroke of the ram. A total of five test bends shall be made, after which the press brake and the test pieces shall be examined. The ram shall be level in accordance with 3.4.2.2. The machine shall not be bowed, sprung, have chipped components, or otherwise be distorted or broken. Each bend shall be clean and uniform throughout. The bends shall be of the correct angle within plus or minus one degree for the full bend length.

4.7.2.2 Repetitive air bending test. The press brake shall then be tested for accuracy in repetitive air bending by making air bends in pieces of sheet or plate 6 inch x 6 inch or larger of the type and strength used in full length bends. The depth stop shall be set to permit a 90° air bend. The depth stop shall not be repositioned or reset during the entire series of tests. A test piece shall be positioned under each end of the ram in alignment with the end housings, and an air bend accomplished. This operation shall be repeated five times. Five additional test pieces shall be bent individually at the left end of the ram with no material under the opposite end. This operation shall be repeated for the right end of the ram. The 20 test peices, when compared, shall not vary more than plus or minus one and one-half degree for sheet up to 10 gauge in thickness and plus or minus two and one-half degrees for plate stock over 10 gauge in thickness.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:34Z
Check the source to verify that this is the current version before use.

4.7.3 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The press brakes covered by this specification are intended for use in forming, bending, flanging, multiple punching, and similar operations on sheet metal and plate.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size required (see 1.2, and table I).

c. First article, if required (see 3.1).

d. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Dual calibrations (US and SI), if required (see 3.2.1.2).

g. Additional safety and health requirements, if required (see 3.2.4).

h. Centralized automatic lubrication system, if required (see 3.2.8).

i. Painting, if different (see 3.3.5).

j. Electrogmagnetic interference control, if required (see 3.3.8)

k. Machined bed, if required (see 3.4.1.1).

Source: https://assist.dla.mil -- Downloaded: 202610-23T04:34Z
Check the source to verify that this is the current version before use.

l. Die holder or filler block, if required (see 3.4.1.1).

m. Type of die and punch, fully describe (see 3.4.1.1).

n. Machined ram, if required (see 3.4.2).

o. Ram tilt adjustment, if different (see 3.4.2.1).

p. Hydraulics conforming to JIC H-1, if required (see 3.4.3).

q. If the hydraulic unit(s) is to be a specific type, state type required (see 3.4.4).

r. Actuator controls, if different (see 3.4.5.2).

s. Ram speed combinations, if different (see 3.4.5.3).

t. Back gauges, if different (see 3.4.6).

u. Electrical system and input power supply, if different (see 3.4.7).

v. Motor enclosure, if different (see 3.4.7.1).

w. Size and capacity, if different (see 3.5).

x. Additional performance features, if required (see 3.6.1).

y. Standard equipment, if different (see 3.7).

z. Specify and fully describe optional equipment required, if different (see 3.8).

aa. Specify applicable repair parts, if required (see 3.9).

bb. Hour meter, if different (see 3.11).

cc. First article inspection, if different (see 4.4).

dd. Quality conformance inspection, if different (see 4.5).

ee. If preservation, packing, and marking in accordance with MIL-M-18058 is required, specify level required (see 5.1).

6.2.2 Contract data requirements. Required technical data such as operator's manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports should be specified on a DD Form 1423, Contract Data Requirements List incorporated into the contract.

Source: https://assist.dla.mil -- Downloaded: 202640-23T04:34Z
Check the source to verify that this is the currentversion before use.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test, and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern the machine in its operating environment, such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. Since OSHA limits the total hazard level of these hazards in the environment and does not limit the hazard level of individual machines in the environment the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional machine requirements that will integrate the new machine into its future operating environment. The machine shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturers' commercial machines supplied to the commercial market. If specific point-of-operation guardng is required, the procuring activity should specify the exact configuration of the guard required, as in most cases, the guard configuration is dependent on the size and configuration of the workpieces. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1 (g).

6.5 Cross-reference of classification changes. Machine classification changes from previous issue of this specification are as follows:

a. The following sizes were deleted from MIL-P-80052B:

12-3	25-8	75-8
12-4	40-6	75-10
25-4	40-8	75-12
25-6	40-10	100-14

b. The following new sizes were added to MIL-P-80052C:

20-4	175-16	350-10	600-10	1250-14	1500-24	2000-22
35-4	225-8	350-12	600-18	1250-22	1500-26	2000-24
35-10	225-10	350-14	600-20	1250-24	1500-28	2000-26
135-8	225-12	350-16	600-22	1250-26	1500-30	2000-28
135-10	225-14	350-18	600-24	1250-28	1500-32	2000-30
135-12	225-16	400-8	750-22	1500-12	1500-34	2000-32
135-14	250-10	400-18	750-24	1500-14	2000-12	2000-34
175-8	250-12	400-20	1000-12	1500-16	2000-14	
175-10	250-14	500-18	1000-22	1500-18	2000-16	
175-12	250-16	500-20	1000-24	1500-20	2000-18	
175-14	350-8	500-24	1250-12	1500-22	2000-20	

Source: https://assist.dla.mil -- Downloaded: 202610-23T04:34Z
Check the source to verify that this is the currentversion before use.

6.6 Subject term (key word) listing.

Angle, bend
Brake, press, forming, bending
Die
Forming, machine
Hydraulic, press
Machine, tool, shop, forming
Metal, fabrication
Plate, steel
Punch
Steel, sheet, plate

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Preparing activity:
DLA-IP

(Project 3441-0112)

Review activities:
Air Force - 84
DLA - GS

User activities:
Army - AR
Navy - YD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:34Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEFENSE LOGISTICS AGENCY
Commander
Defense Industrial Plant Equipment Center
ATTN: DIPEC-SSG
Memphis, TN 38114-5297

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 4966 WASHINGTON D. C

POSTAGE WILL BE PAID BY THE DEFENSE LOGISTICS AGENCY

Commander
Defense Industrial Plant Equipment Center
ATTN: DIPEC-SSG
Memphis, TN 38114-5297

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:34Z
Check the source to verify that this is the current version before use.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 202610-23T04:34Z
Check the source to verify that this is the current version before use.